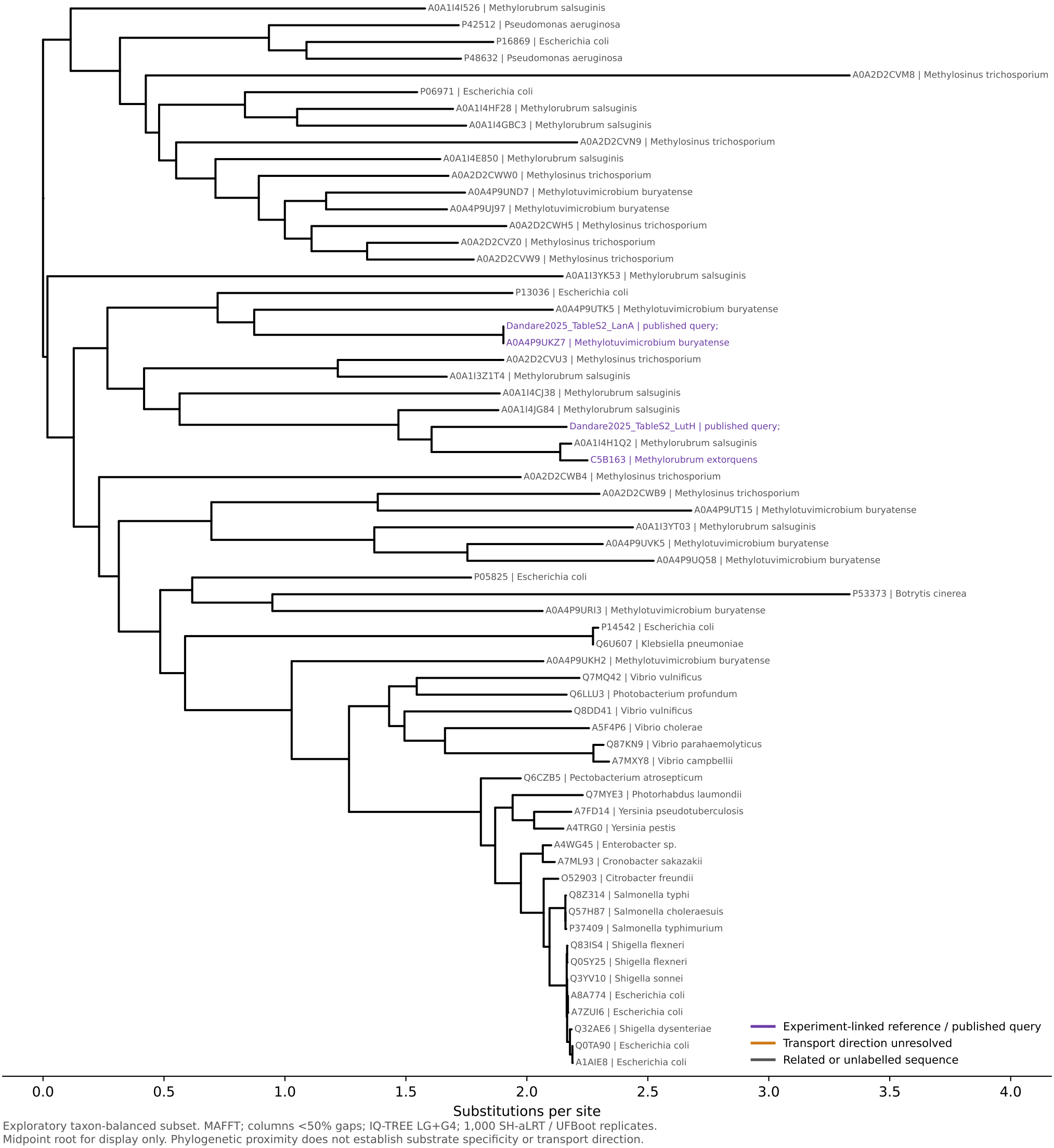
Lanthanide-receptor references and related TonB proteins



Exploratory taxon-balanced subset. MAFFT; columns <50% gaps; IQ-TREE LG+G4; 1,000 SH-aLRT / UFBoot replicates.
Midpoint root for display only. Phylogenetic proximity does not establish substrate specificity or transport direction.